

Matter-Conditioned Metric Response with Coherence-Weighted Irreversibility: A Galaxy-Held-Out CREST Test on SPARC

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Abstract

This paper rewrites CREST (Covariant Relativistic Extended Scalar Theory) as a scalar-tensor metric-response framework with an explicit coherence-weighted irreversibility law. The response sector is defined covariantly by the residual curvature tensor

$$E_{\mu\nu}^{\chi} \equiv G_{\mu\nu} - 8\pi G T_{\mu\nu}^{\text{bar}},$$

and becomes predictive only after a constrained response law is declared,

$$E_{\mu\nu}^{\chi} = A_{\mu\nu}[\chi, \nabla\chi, \nabla\nabla\chi, \eta, I, F, \Phi_{\text{irr}}, R_{\chi}, S_{\chi}, g_{\mu\nu}].$$

The core upgrade is the separation of integrated irreversible response load F from instantaneous irreversible response flux Φ_{irr} . The static admissibility condition is still governed by a degraded response reserve,

$$\chi^2 + R_{\chi} = \eta I e^{-\lambda_F F},$$

but the dynamical pressure on the response sector is now measured by the coherence-weighted process-time load

$$\Theta_{\chi} = \frac{\Phi_{\text{irr}}}{\chi \eta^2 I + \epsilon}, \quad \frac{d\tau_{\chi}}{dt} = \kappa_{\chi} \Theta_{\chi}^p.$$

Thus matter conditions the metric response, coherent informational capacity stabilizes it, and irreversible flux advances response-sector history, branch erosion, and residual broadening.

The empirical channel is deliberately narrow: galaxy-held-out prediction on SPARC rotation-curve mass models. The benchmark uses 3384 pointwise measurements across 175 galaxies after finite-value checks. A declared CREST v5b response projector improves median galaxy RMSE by 7.33% over a fitted RAR/MOND-like backbone and by 3.54% over a radius-plus-baryonic-acceleration nuisance control. It also outperforms no-retuning NFW/Burkert halo-prior predictors, while flexible per-galaxy halo fits remain stronger raw curve-fitting ceilings. Null-collapse tests show that the gain is not reproduced when the organized invariant-sector alignment is destroyed inside matched nuisance bins. The claim is therefore conservative: CREST currently provides a falsifiable, no-retuning metric-response projector advantage in SPARC residual prediction. It is not yet a replacement for dark matter, a lensing/cosmology closure, or a unique microscopic action.

Keywords: galaxy rotation curves; SPARC; radial acceleration relation; scalar-tensor gravity; metric response; irreversible flux; process time; dark matter alternatives; held-out prediction.

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1 Purpose and claim boundary

CREST stands for *Covariant Relativistic Extended Scalar Theory*. In its most precise physics-facing form, it is a scalar-tensor metric-response framework: the spacetime metric remains the tensorial gravitational variable, while χ denotes a scalar projection of an extended matter-conditioned response sector.

The earlier galaxy-facing formulation already separated the observed circular acceleration into a baryonic part and an excess response part,

$$g_{\text{obs}}(R) = g_{\text{bar}}(R) + g_{\chi}(R). \quad (1)$$

That is useful phenomenologically, but it is not by itself enough. If g_{χ} is treated only as a residual, the framework risks becoming a renamed halo fit. If χ is described only as support or backbone language, it risks sounding metaphorical. The goal of this rewrite is to make the physical division of labor explicit:

χ is not hidden particulate matter, not empty space, and not a metaphysical substrate. It is the projected scalar amplitude of a matter-conditioned metric response. The load sector is not generic scatter; it is irreversible response flux and its accumulated history, normalized by coherent informational capacity.

The paper therefore has two connected tasks. First, it gives the response sector a covariant definition compatible with scalar-tensor reasoning and Bianchi consistency. Second, it uses the coherence-weighted process-time law to sharpen the role of irreversible loading in the galaxy projector.

1.1 Positive claim

The positive claim is:

A framework-constrained CREST response projector improves galaxy-held-out SPARC prediction over a fitted RAR/MOND-like baseline, over radius-plus- g_{bar} nuisance controls, and over no-retuning NFW/Burkert halo-prior predictors. The improvement collapses under matched null actions that destroy invariant-sector alignment while preserving nuisance structure. The coherence-weighted irreversibility law explains why irreversible load must be normalized by coherent informational capacity rather than treated as free scatter.

1.2 Non-claims

This paper does not claim that:

1. particulate dark matter has been disproven;
2. the final relativistic response action is unique;
3. SPARC alone proves lensing, CMB, cluster, structure-growth, or cosmological closure;
4. flexible per-galaxy halo fits are beaten in raw curve-fitting capacity;

5. Q1-only evidence is independently decisive under full-refit nulls;
6. coherence-weighted process time replaces relativistic proper time.

These boundaries are part of the result. They keep the metric-response sector falsifiable and prevent $A_{\mu\nu}$ from becoming a universal residual label.

2 Notation and invariant sector

A domain-specific state $x \in \mathcal{X}$ is mapped to a response-state vector by a projector

$$\Pi_{\chi} : x \mapsto z = (\chi, \eta, I, F, \Phi_{\text{irr}}). \quad (2)$$

The quantities used here are summarized in Table 1.

Table 1: Response-sector notation.

Symbol	Name	Interpretation
χ	scalar response amplitude	projected scalar amplitude of matter-conditioned metric response
η	ordered response fraction	fraction of the response organized coherently rather than incoherently or contaminatingly
I	structural information	coarse-grained structural information stored in the baryon-response configuration
F	integrated irreversible load	accumulated response-sector degradation, disequilibrium history, or irreversible loading
Φ_{irr}	irreversible response flux	instantaneous nonrecoverable response update, entropy-production proxy, kinematic disequilibrium flux, or record-production rate
R_{χ}	response reserve	remaining degraded admissibility margin
S_{χ}	support-boundary coordinate	logarithmic distance from response saturation
Θ_{χ}	process-time load	irreversible flux per coherent informational response capacity

The key correction in this rewrite is the separation

$$F(t) = F(t_0) + \int_{t_0}^t \Phi_{\text{irr}}(t') dt', \quad (3)$$

with possible coarse-graining factors absorbed into Φ_{irr} . The old F language mixed instantaneous disequilibrium and accumulated degradation. The upgraded theory keeps both: F gates the remaining reserve, while Φ_{irr} drives the rate at which the sector moves through irreversible response history.

3 Covariant definition of metric response

3.1 Einstein residual

Let $G_{\mu\nu}$ be the Einstein tensor and $T_{\mu\nu}^{\text{bar}}$ the baryonic stress-energy tensor. The diagnostic residual curvature response is

$$E_{\mu\nu}^{\chi} \equiv G_{\mu\nu} - 8\pi G T_{\mu\nu}^{\text{bar}}. \quad (4)$$

Equation (4) is an identity, not yet a theory. CREST becomes predictive only when the residual is constrained by a response law,

$$E_{\mu\nu}^{\chi} = A_{\mu\nu}[\chi, \nabla\chi, \nabla\nabla\chi, \eta, I, F, \Phi_{\text{irr}}, R_{\chi}, S_{\chi}, g_{\mu\nu}], \quad (5)$$

so that

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{bar}} + A_{\mu\nu}. \quad (6)$$

The tensor $A_{\mu\nu}$ is not particulate dark matter. It may be moved to the right-hand side and written as an effective stress tensor,

$$T_{\chi, \mu\nu}^{\text{eff}} = \frac{1}{8\pi G} A_{\mu\nu}, \quad (7)$$

but that is bookkeeping, not an assumption that the sector consists of hidden particles.

3.2 Action-level origin and Bianchi gate

A macroscopic response action can be written schematically as

$$S = S_{\text{EH}} + S_{\text{bar}} + S_{\text{resp}}, \quad (8)$$

with

$$S_{\text{resp}} = \int d^4x \sqrt{-g} \left[\mathcal{L}_{\text{dyn}}(\chi, \eta, I, F, \nabla\chi, \dots) + \Lambda \left(\eta I e^{-\lambda_F F} - \chi^2 - R_{\chi} \right) \right]. \quad (9)$$

The Lagrange multiplier Λ enforces the degraded reserve law. Metric variation defines

$$A_{\mu\nu} \equiv -\frac{16\pi G}{\sqrt{-g}} \frac{\delta S_{\text{resp}}}{\delta g^{\mu\nu}}. \quad (10)$$

Diffeomorphism invariance imposes

$$\nabla^{\mu} \left(8\pi G T_{\mu\nu}^{\text{bar}} + A_{\mu\nu} \right) = 0. \quad (11)$$

If baryonic stress-energy is separately conserved, then $\nabla^{\mu} A_{\mu\nu} = 0$. If baryons exchange energy-momentum with the response sector, the exchange current must close Eq. (11). This is the relativistic consistency gate.

4 State-side support and degraded response reserve

The state-side support matrix is

$$M_A = \begin{pmatrix} \eta & \chi \\ \chi & I \end{pmatrix}. \quad (12)$$

For $\eta \geq 0$ and $I \geq 0$, positive semidefiniteness is equivalent to

$$M_A \succeq 0 \iff R_A \equiv \eta I - \chi^2 \geq 0. \quad (13)$$

The support reserve R_A is the undegraded state-side admissibility margin. In the metric-response sector, irreversible load reduces active capacity. The degraded reserve law is

$$\chi^2 + R_\chi = \eta I e^{-\lambda_F F}, \quad (14)$$

or

$$R_\chi = \eta I e^{-\lambda_F F} - \chi^2, \quad \chi^2 \leq \eta I e^{-\lambda_F F}. \quad (15)$$

A logarithmic boundary coordinate is

$$S_\chi = \ln \left(\frac{\eta I e^{-\lambda_F F}}{\chi^2 + \epsilon_\chi} \right). \quad (16)$$

The saturation surface is $S_\chi = 0$. In empirical projectors, F may gate and degrade organized response capacity, but it is not permitted to become a free residual absorber.

5 Coherence-weighted process time

5.1 Capacity

The coherence-weighted informational response capacity is

$$C_\chi = \chi \eta^2 I. \quad (17)$$

The factor I measures irreducible structural information, η^2 encodes the pairwise or relation-like character of coherent organization, and χ measures persistence of the response channel. A generalized family is possible,

$$C_\chi(a, b, c) = \chi^a \eta^b I^c, \quad a, b, c > 0, \quad (18)$$

but the minimal default used in this paper is $a = 1, b = 2, c = 1$.

5.2 Process-time law

Let $d\Sigma_{\text{irr}}$ denote an irreversible response-record increment. In a galaxy application it may be approximated by dissipative disequilibrium, non-circular kinetic complexity, uncertainty-weighted response turnover, or other projector-specific irreversible load proxies. The process-time law is

$$d\tau_\chi = \kappa_\chi \frac{d\Sigma_{\text{irr}}}{C_\chi + \epsilon}. \quad (19)$$

If $d\Sigma_{\text{irr}}/dt = \Phi_{\text{irr}}$, then

$$\frac{d\tau_\chi}{dt} = \kappa_\chi \left(\frac{\Phi_{\text{irr}}}{\chi \eta^2 I + \epsilon} \right)^p, \quad p > 0. \quad (20)$$

The default model takes $p = 1$ until data require nonlinear scaling.

Define the process-time load index

$$\Theta_\chi \equiv \frac{\Phi_{\text{irr}}}{\chi \eta^2 I + \epsilon}. \quad (21)$$

Large Θ_χ means high irreversible response pressure: rapid residual broadening, response-branch erosion, kinematic disequilibrium, or increased event turnover. Small Θ_χ means the same flux is hosted by larger coherent informational capacity.

5.3 Loaded reserve with time-pressure gating

The simplest static law keeps Eq. (14). A stronger dynamical closure allows the reserve to depend on both accumulated load and process time:

$$\chi^2 + R_\chi = \eta I \exp[-\lambda_F F - \lambda_\tau \tau_\chi]. \quad (22)$$

Equation (22) is not used as an unconstrained fit in the present SPARC benchmark. It is the theory-facing upgrade: F measures accumulated load, while τ_χ measures how fast irreversible history advances relative to coherent capacity. The galaxy projector below implements this principle conservatively through load-gated and capacity-normalized features.

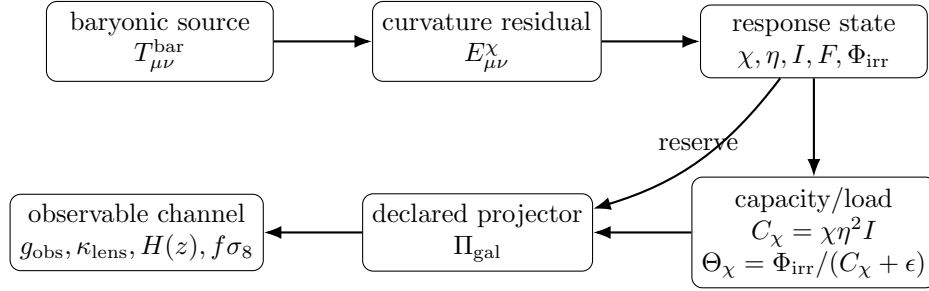


Figure 1: CREST response architecture after the coherence-weighted upgrade. The response sector is covariant at the tensor level but empirical only through declared projectors.

6 Weak-field galaxy projection

For a weak-field, quasi-static galaxy, write

$$ds^2 \simeq - \left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Psi}{c^2}\right) d\mathbf{x}^2. \quad (23)$$

The potentials split into baryonic and response components,

$$\Phi = \Phi_{\text{bar}} + \Phi_\chi, \quad \Psi = \Psi_{\text{bar}} + \Psi_\chi. \quad (24)$$

The circular acceleration channel is

$$g_{\text{obs}}(R) = g_{\text{bar}}(R) + g_\chi(R), \quad g_\chi(R) = |\nabla \Phi_\chi(R)|. \quad (25)$$

A scalar galaxy projection may be written as

$$\chi_g^2(R) = \frac{g_\chi(R)}{g_*}, \quad (26)$$

so the degraded reserve gives a weak-field closure

$$g_\chi(R) = g_* \left[\eta(R) I(R) e^{-\lambda_F F(R)} - R_\chi(R) \right]. \quad (27)$$

The empirical model does not fit Eq. (27) freely at every radius. It builds a low-dimensional residual projector constrained by the division of labor among χ , η , I , F , and Φ_{irr} .

7 Projector discipline

A scalar projector from invariant state to observable channel has the form

$$\Pi_{D,h}(z) = W_{D,h} \left(\frac{E_{D,h}(R_D z)}{K_{D,h}(R_D z)} \right), \quad (28)$$

where R_D is the domain representation map, $E_{D,h}$ is an organized-effect functional, $K_{D,h}$ is a capacity or hosting normalization, and $W_{D,h}$ maps the quotient into the observable codomain.

For galaxy residuals,

$$\log_{10} g_{\text{obs}}(R) = \log_{10} g_{\text{RAR}}(R) + \Pi_{\text{gal}}(R) + \epsilon(R). \quad (29)$$

The projection layer enforces four rules:

1. no galaxy may appear in both train and test folds;
2. a radius/ g_{bar} nuisance control must be beaten;
3. no-retuning halo-prior predictors must be compared separately from per-galaxy halo ceilings;
4. null actions must destroy organized invariant-sector alignment while preserving nuisance structure.

The coherence-weighted upgrade adds a fifth rule:

5. irreversible load must be represented through capacity-normalized pressure Θ_χ or a declared proxy, not as an arbitrary scatter sink.

8 Data and reconstruction

The tests use the SPARC archive of disk galaxies with Spitzer photometry and resolved rotation-curve mass models. The pointwise mass-model table supplies galaxy ID, distance, radius R , observed velocity V_{obs} , velocity uncertainty, gas contribution, disk contribution, bulge contribution, disk surface brightness, and bulge surface brightness. The galaxy table supplies distance, inclination, luminosity, effective radius, disk scale length, H I mass, H I radius, flat velocity, quality flag, and references. Quality flags are used as $Q = 1$ high quality, $Q = 2$ medium quality, and $Q = 3$ low quality.

The baryonic velocity contribution is reconstructed as

$$V_{\text{bar}}^2(R) = V_{\text{gas}}^2(R) + \Upsilon_d V_{\text{disk}}^2(R) + \Upsilon_b V_{\text{bul}}^2(R), \quad (30)$$

with the main analysis using fixed $\Upsilon_d = \Upsilon_b = 0.5$ at $3.6 \mu\text{m}$, matching the standard SPARC/RAR convention. Accelerations are

$$g_{\text{bar}}(R) = \frac{V_{\text{bar}}^2(R)}{R}, \quad g_{\text{obs}}(R) = \frac{V_{\text{obs}}^2(R)}{R}. \quad (31)$$

After quality and finite-value checks the benchmark contains 3384 pointwise measurements across 175 galaxies.

9 Baselines and held-out protocol

9.1 RAR/MOND-like backbone

The RAR/MOND-like baseline is

$$g_{\text{RAR}} = \nu \left(\frac{g_{\text{bar}}}{g_{\dagger}} \right) g_{\text{bar}}, \quad \nu(y) = \frac{1}{1 - e^{-\sqrt{y}}}. \quad (32)$$

The acceleration scale $g_{\dagger}$ is fitted on training galaxies only in each grouped fold. The residual being modeled is

$$r(R) = \log_{10} g_{\text{obs}}(R) - \log_{10} g_{\text{RAR}}(R). \quad (33)$$

9.2 Radius and baryonic-acceleration nuisance control

The nuisance model uses only generic radial and baryonic-acceleration coordinates,

$$X_{\text{nuis}} = \{z, z^2, z^3, \ell_g, \ell_g^2, z\ell_g\}, \quad z = \ln(R/R_d), \quad \ell_g = \log_{10} g_{\text{bar}}. \quad (34)$$

This answers the leakage question: does CREST add information beyond generic curve-shape dependence on radius and baryonic acceleration?

9.3 Coherence-weighted CREST v5b projector

The headline projector is a framework-shrinkage model. Smoothed structural features are computed per galaxy from regularized derivatives of $\ln \Sigma_{\text{disk}}$, $\ln g_{\text{bar}}$, and the RAR response excess. The structural information sectors include

$$I_{\text{grad}}(R) \sim \left(\frac{d \ln \Sigma}{d \ln R} \right)^2 + \left(\frac{d \ln g_{\text{bar}}}{d \ln R} \right)^2 + \left(\frac{d \ln g_{\chi}}{d \ln R} \right)^2, \quad (35)$$

$$I_{\text{curv}}(R) \sim \left(\frac{d^2 \ln \Sigma}{d(\ln R)^2} \right)^2 + \left(\frac{d^2 \ln g_{\text{bar}}}{d(\ln R)^2} \right)^2 + \left(\frac{d^2 \ln g_{\chi}}{d(\ln R)^2} \right)^2. \quad (36)$$

The ordered fraction proxy is a decreasing function of curvature, mixed structural variation, and local sampling spacing,

$$\eta_{\text{struct}} = \frac{1}{1 + 0.35 I_{\text{curv}} + 0.15 I_{\text{mix}} + 0.10 \Delta x_{\text{local}}}. \quad (37)$$

The clean and load-degraded capacities are

$$C_{\text{load}} = \eta_{\text{struct}} (1 + I_{\text{struct}}) e^{-\lambda F_{\text{total}}}, \quad C_{\text{clean}} = \eta_{\text{struct}} (1 + I_{\text{struct}}) e^{-\lambda F_{\text{sampling}}}, \quad (38)$$

with $\lambda = 0.75$ and

$$F_{\text{total}} = F_{\text{sampling}} + F_{\text{error}} + F_{\text{obs unc}}. \quad (39)$$

The internal projector quotient is

$$Q_{\text{CREST}}(R) = \frac{\eta_{\text{struct}} (I_{\text{grad}} + 0.75 I_{\text{curv}} + 0.25 I_{\text{mix}}) e^{-\lambda F}}{1 + \chi^2 + 0.25 |z| + 0.1 Q_{\text{bar}}}. \quad (40)$$

The coherence-weighted interpretation of Eq. (40) is that the numerator measures organized structural response after load degradation, while the denominator prevents the correction from growing merely because the radius or backbone residual is large. The upgraded variable to test in the next-generation projector is

$$\Theta_{\chi,\text{gal}}(R) = \frac{\hat{\Phi}_{\text{irr}}(R)}{\hat{\chi}(R)\hat{\eta}(R)^2\hat{I}(R) + \epsilon}, \quad (41)$$

with $\hat{\Phi}_{\text{irr}}$ estimated from sampling load, uncertainty load, kinematic asymmetry proxies, non-circular residual stress, or two-dimensional H I velocity-field diagnostics when available.

All continuous features are clipped according to training-fold quantiles before test prediction. Let $\hat{r}_{\text{nuis}}$ be the nuisance residual prediction and $\hat{r}_{\text{load}}$ the load-gated framework residual prediction. The final v5b correction is

$$\hat{r}_{\text{v5b}} = \hat{r}_{\text{nuis}} + \alpha(\hat{r}_{\text{load}} - \hat{r}_{\text{nuis}}). \quad (42)$$

The scalar shrinkage coefficient $\alpha \in [0, 1]$ is selected by inner grouped cross-validation on $Q = 1 + 2$ training galaxies only, then frozen for the outer held-out fold. This is the practical implementation of the rule that load gates or degrades response but does not become an arbitrary residual field.

9.4 Halo-prior and halo-ceiling comparisons

No-retuning NFW and Burkert halo-prior predictors are trained on other galaxies and use baryonic/structural metadata to predict held-out halo-shape behavior. They do not fit the held-out galaxy rotation curve. Per-galaxy NFW and Burkert ceiling models are also reported, but they are a different comparison: they are allowed to fit halo parameters directly to the held-out curve, so they measure curve-fitting capacity rather than no-retuning prediction.

10 Results

10.1 Framework projector versus RAR and nuisance controls

Table 2 gives the headline v5b comparison. The framework-shrinkage model improves median galaxy RMSE over RAR by 7.33% and over the radius/ g_{bar} nuisance control by 3.54%. The surrogate framework correction remains close to the nuisance model, indicating that marginal feature information alone is insufficient.

Table 2: Headline v5b galaxy-held-out performance. Lower RMSE is better. Improvements are computed from median galaxy RMSE.

Model	Med. gal. RMSE	Mean gal. RMSE	Pt.-weighted RMSE	vs RAR (%)	vs control (%)
RAR	0.1520	0.1940	0.1428	0.00	-4.09
radius + g_{bar} control	0.1461	0.1928	0.1407	3.93	0.00
CREST load diagnostic	0.1452	0.1869	0.1389	4.52	0.62
CREST v5b	0.1409	0.1833	0.1382	7.33	3.54
surrogate CREST	0.1450	0.1933	0.1414	4.65	0.75

10.2 Quality-stratified behavior

The all-galaxy and $Q = 1 + 2$ subsets are the main publication-facing subsets. The $Q = 1$ subset is positive relative to the nuisance control but not positive relative to RAR in median RMSE; this limitation is important for the claim boundary.

Table 3: Quality-stratified v5b performance. The $Q = 1 + 2$ subset is the main clean-plus-medium validation class.

Subset	N_{gal}	Med. RMSE	vs RAR (%)	vs control (%)
Q1	99	0.1302	-1.30	2.26
Q1+Q2	163	0.1327	5.44	3.80
Q2	64	0.1489	12.24	8.09
Q3	12	0.4588	10.01	7.41
all	175	0.1409	7.33	3.54

10.3 No-retuning halo priors and flexible halo ceilings

Table 4 separates no-retuning halo-prior prediction from flexible halo-fitting capacity. The CREST v5b projector outperforms all no-retuning NFW/Burkert prior variants tested here. Flexible halo ceilings substantially outperform CREST because they are fitted directly to each held-out galaxy curve.

Table 4: Halo-prior and halo-ceiling comparison. Positive values in the last columns mean the listed model improves over the named baseline; negative values mean it is worse.

Model	Med. gal. RMSE	vs RAR (%)	vs control (%)	vs CREST (%)
RAR	0.1520	0.00	-4.09	-7.90
radius + g_{bar} control	0.1461	3.93	0.00	-3.66
CREST v5b	0.1409	7.33	3.54	0.00
NFW rich no-retuning prior	0.1630	-7.24	-11.63	-15.72
Burkert rich no-retuning prior	0.1643	-8.04	-12.46	-16.58
NFW mass/size prior	0.1661	-9.22	-13.69	-17.86
Burkert mass/size prior	0.1672	-10.00	-14.50	-18.70
NFW mass-only prior	0.1674	-10.09	-14.59	-18.79
Burkert mass-only prior	0.1586	-4.33	-8.60	-12.58
NFW 1-param ceiling	0.0902	40.65	38.23	35.96
Burkert 1-param ceiling	0.0934	38.59	36.07	33.73
NFW 2-param ceiling	0.0776	48.96	46.87	44.92
Burkert 2-param ceiling	0.0564	62.90	61.38	59.97

10.4 Null-collapse tests

A projector that only fits better is not enough. It must collapse when the claimed organized structure is destroyed while nuisance statistics are preserved. Two null stages were run. The first was a 1000-replicate post-fit null: it shuffles or rolls the already-fit CREST correction inside matched nuisance bins while preserving Q , g_{bar} , and R/R_d structure. The second was a stricter but smaller

full-refit null with 30 replicates per null mode: it refits the load-sector model after matched feature or row shuffles.

Table 5 summarizes the null-collapse result. The post-fit nulls strongly reject nuisance leakage for all galaxies and for $Q = 1 + 2$. The lightweight full-refit null also rejects for all galaxies and $Q = 1 + 2$. The $Q = 1$ result remains positive but not decisive under full-refit nulls.

Table 5: Null-collapse summary. The v9 column reports the most conservative post-fit null p -value among the tested null modes. The v10 column reports the most conservative lightweight full-refit null p -value.

Subset	Real imp. (%)	v9 worst $p_{\text{null} \geq \text{real}}$	v10 worst $p_{\text{null} \geq \text{real}}$	v10 null 95th max (%)
all	3.54	0.000	0.000	-0.16
Q1+Q2	3.80	0.000	0.000	0.15
Q1	2.26	0.041	0.500	4.49
Q3	7.41	0.033	0.000	1.00

11 What the coherence law adds

The coherence-weighted law improves the CREST paper in four concrete ways.

11.1 It separates admissibility from history

The reserve inequality

$$\chi^2 \leq \eta I e^{-\lambda_F F} \quad (43)$$

answers a state question: is the response amplitude admissible given the ordered structural capacity left after degradation? The process-time law

$$\frac{d\tau_\chi}{dt} = \kappa_\chi \left(\frac{\Phi_{\text{irr}}}{\chi \eta^2 I + \epsilon} \right)^p \quad (44)$$

answers a history question: how quickly does the sector generate irreversible response history per unit coherent capacity? These are not the same. A galaxy can be inside the admissible region but under high irreversible time pressure. Conversely, a galaxy can have substantial response reserve but little current flux.

11.2 It prevents F from becoming a scatter label

If F is merely inserted as a damping factor, it can look like a flexible residual knob. The upgraded theory makes the load sector measurable: one must declare Φ_{irr} proxies, capacity estimates, and nulls. A load correction is acceptable only if it improves prediction beyond nuisance coordinates and collapses when organized alignment is destroyed.

11.3 It predicts branch broadening

The next clean empirical prediction is not just a lower RMSE. It is a conditional branch-thickness statement:

$$\text{Var}[r_{\text{gal}} | g_{\text{bar}}, R/R_d, Q] \text{ should increase with } \Theta_{\chi, \text{gal}}, \quad (45)$$

where $r_{\text{gal}} = \log_{10} g_{\text{obs}} - \log_{10} g_{\text{RAR}}$ and $\Theta_{\chi, \text{gal}}$ is estimated from coherence-weighted irreversible load proxies. This is stronger than saying contaminated galaxies scatter more. It says that scatter should follow irreversible load normalized by coherent structural capacity.

11.4 It connects galaxy dynamics to other projectors

The same parent response sector should project into lensing, redshift, timing, cosmology, and growth. Therefore the galaxy improvement cannot be treated as final closure. The coherence law gives these channels a common language: each must estimate irreversible flux, coherent capacity, and the observable projector mapping from $A_{\mu\nu}$ to its data product.

12 Lensing, time, and cosmological projections

12.1 Lensing and gravitational slip

Rotation curves mainly constrain Φ_χ through $g_\chi = |\nabla \Phi_\chi|$. Lensing responds to the Weyl combination,

$$\Phi_{\text{lens}} \propto \Phi + \Psi. \quad (46)$$

Thus CREST must declare a slip relation,

$$\Psi_\chi = \Gamma_\chi[\chi, \eta, I, F, \Phi_{\text{irr}}]\Phi_\chi, \quad (47)$$

and test whether the same response field that helps rotation curves also produces acceptable weak-lensing and galaxy-galaxy lensing signals. If $\Gamma_\chi = 1$, the leading lensing response is GR-like for the effective metric response. If not, the slip must be fit or derived and then constrained by lensing data.

12.2 Time and redshift

The process-time variable τ_χ is not proper time. Relativistic proper time remains determined by the metric,

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu. \quad (48)$$

The role of τ_χ is internal response history: irreversible branch updating, degradation, state turnover, or response broadening. A time/redshift projector must therefore distinguish metric-clock effects from response-history effects. A falsifiable prediction is that regions with high Θ_χ should show larger response-sector broadening or relaxation effects after controlling for the ordinary metric potential.

12.3 Cosmological background and growth

At the background level one may represent the homogeneous response contribution as an effective density and pressure,

$$\rho_\chi(a), \quad p_\chi(a) = w_\chi(a)\rho_\chi(a), \quad (49)$$

with a response equation of state constrained by the same invariant sector. A conservative phenomenological prototype is

$$w_\chi(a) = w_0 + w_a(1 - a) + \frac{\beta_\chi a^2}{a^2 + a_\chi^2}, \quad (50)$$

but the coherence-weighted upgrade says this is not enough. A complete cosmology must connect $w_\chi(a)$ and perturbation behavior to F , Φ_{irr} , C_χ , and the covariant response tensor $A_{\mu\nu}$. The required tests are CMB, BAO, supernovae, growth, and weak-lensing summaries under a single Boltzmann-code implementation.

13 Discussion

13.1 What the result supports

The analysis supports a narrow but useful claim: the CREST projector contains held-out predictive information beyond a fitted RAR backbone and beyond generic radius/ g_{bar} curve-shape nuisance terms. The signal also survives comparisons with no-retuning NFW/Burkert halo-prior predictors and collapses under matched nulls. This is the empirical behavior expected if invariant response-sector features carry organized residual information.

The coherence-weighted law strengthens the interpretation by showing why irreversible load must be normalized by coherent informational capacity. A high-load galaxy is not automatically a strong CREST signal. A high-load galaxy with low coherent capacity should have high Θ_χ and therefore greater response-history pressure; a high-load galaxy with high coherent capacity may remain comparatively stable.

13.2 What the result does not support

The analysis does not show that dark matter is false. It does not show that fully tuned halos are beaten; the per-galaxy halo ceilings are much better at raw curve fitting. It also does not establish relativistic lensing or cosmological closure. The correct interpretation is prediction efficiency: a constrained response-sector projector predicts held-out galaxy residual structure with fewer per-galaxy freedoms than flexible halo fits.

13.3 Why Q1 alone is not the headline

The $Q = 1$ subset is high quality and therefore important. In v5b it improves over the nuisance control by 2.26%, but it does not improve over RAR in median RMSE, and the lightweight full-refit null overlaps the real improvement. This prevents a stronger claim based on Q1 alone. The publication-facing validation class should therefore be $Q = 1 + 2$, with Q1 reported transparently as positive versus nuisance but not independently decisive.

13.4 Why Q3 is not the headline

The low-quality $Q = 3$ subset shows large improvements, but such galaxies are more vulnerable to non-circular motion, sampling irregularity, inclination uncertainty, and other observational complications. The $Q = 3$ result is useful as a stress test and as a clue about disequilibrium sensitivity, not as the evidential center of the paper.

14 Falsifiability and next tests

The galaxy-sector claim fails or weakens if any of the following occur:

1. larger full-refit null runs eliminate the all-galaxy or $Q = 1 + 2$ advantage;
2. an improved no-retuning halo-prior model beats v5b under the same held-out protocol;
3. the CREST advantage disappears under independently written code;
4. $\Theta_{\chi, \text{gal}}$ fails to predict residual branch broadening or kinematic disequilibrium after nuisance controls;
5. the same response potential cannot produce acceptable lensing in the leading $\Psi_\chi = \Phi_\chi$ channel;
6. time-delay, redshift, background expansion, or growth projections become inconsistent with the response needed for galaxies.

The most important next tests are:

1. larger full-refit nulls with 500–1000 refits per null mode;
2. hierarchical CREST versus hierarchical halo priors under matched effective degrees of freedom;
3. two-dimensional H I velocity-field stress tests to estimate $\hat{\Phi}_{\text{irr}}$ directly;
4. weak-lensing and galaxy-galaxy lensing projections from the same Φ_χ, Ψ_χ sector;
5. full Boltzmann-code cosmology with CMB, BAO, supernovae, growth, and weak-lensing summaries;
6. an action-level stability analysis checking ghosts, gradient instabilities, superluminal modes, and well-posedness.

15 Conclusion

This paper presents a self-contained CREST rewrite incorporating the coherence-weighted process-time law. The theoretical core is the identification of χ as a scalar projection of a matter-conditioned metric-response tensor, not as a hidden matter species. The degraded reserve law

$$\chi^2 + R_\chi = \eta I e^{-\lambda_F F} \quad (51)$$

separates active response amplitude, ordered response fraction, structural information, accumulated irreversible degradation, and remaining response reserve. The coherence-weighted law

$$\Theta_\chi = \frac{\Phi_{\text{irr}}}{\chi \eta^2 I + \epsilon}, \quad \frac{d\tau_\chi}{dt} = \kappa_\chi \Theta_\chi^p \quad (52)$$

adds the missing dynamical layer: irreversible flux advances response-sector history only relative to the coherent informational capacity available to host it.

The empirical core is the v5b framework-shrinkage projector. Under galaxy-held-out validation it improves median galaxy RMSE by 7.33% over RAR and by 3.54% over a radius/ g_{bar} nuisance control across all SPARC galaxies. It improves the main $Q = 1 + 2$ validation subset by 3.80% over the nuisance control and beats no-retuning NFW/Burkert halo-prior predictors. The null-collapse tests show that the improvement is not reproduced when invariant-sector alignment is destroyed inside matched nuisance bins, especially for all galaxies and for $Q = 1 + 2$.

The strongest conclusion remains conservative: CREST currently provides a constrained, falsifiable, no-retuning response-projector advantage in SPARC residual prediction. The coherence-weighted upgrade makes that advantage more physically disciplined, but the framework still requires lensing closure, cosmological closure, larger nulls, and an action-level stability analysis before it can claim to replace dark matter.

A Compact mathematical summary

The core equations are:

$$E_{\mu\nu}^{\chi} = G_{\mu\nu} - 8\pi G T_{\mu\nu}^{\text{bar}}, \quad (53)$$

$$E_{\mu\nu}^{\chi} = A_{\mu\nu}[\chi, \nabla\chi, \nabla\nabla\chi, \eta, I, F, \Phi_{\text{irr}}, R_{\chi}, S_{\chi}, g_{\mu\nu}], \quad (54)$$

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{bar}} + A_{\mu\nu}, \quad (55)$$

$$\chi^2 + R_{\chi} = \eta I e^{-\lambda_F F}, \quad (56)$$

$$S_{\chi} = \ln \left(\frac{\eta I e^{-\lambda_F F}}{\chi^2 + \epsilon_{\chi}} \right), \quad (57)$$

$$F(t) = F(t_0) + \int_{t_0}^t \Phi_{\text{irr}}(t') dt', \quad (58)$$

$$C_{\chi} = \chi \eta^2 I, \quad (59)$$

$$\Theta_{\chi} = \frac{\Phi_{\text{irr}}}{C_{\chi} + \epsilon}, \quad (60)$$

$$\frac{d\tau_{\chi}}{dt} = \kappa_{\chi} \Theta_{\chi}^p, \quad (61)$$

$$g_{\text{obs}}(R) = g_{\text{bar}}(R) + g_{\chi}(R), \quad (62)$$

$$\log_{10} g_{\text{obs}}(R) = \log_{10} g_{\text{RAR}}(R) + \Pi_{\text{gal}}(R) + \epsilon(R). \quad (63)$$

B Reference implementation checklist

A reproducible implementation should store the following artifacts:

- SPARC galaxy table and mass-model table used for the run;
- reconstruction script for V_{bar} , g_{bar} , and g_{obs} ;
- grouped train/test split definitions by galaxy;

- RAR fitting script with training-fold-only $g_{\dagger}$;
- nuisance-control script using Eq. (34);
- CREST feature-generation script for I_{grad} , I_{curv} , η_{struct} , load proxies, and Q_{CREST} ;
- inner grouped cross-validation script selecting α ;
- no-retuning halo-prior comparison scripts;
- per-galaxy halo-ceiling scripts clearly separated from no-retuning predictors;
- null-collapse scripts for post-fit and full-refit nulls;
- CSV summaries containing pointwise RMSE, galaxy RMSE, subset RMSE, and null distributions;
- next-generation $\Theta_{\chi, \text{gal}}$ branch-broadening test.

C Data availability

The analysis is designed around public SPARC data products associated with Lelli, McGaugh, and Schombert, including the galaxy sample, mass models, RAR table, and BTFR comparison products. A standard public SPARC source page is <https://astroweb.cwru.edu/SPARC/>. The local computation archive for the present paper includes the exact scripts and generated CSV summaries for the v5b, halo-prior, post-fit null, and full-refit null layers.

D Boundary language for reuse

Use the following language when describing the result:

A framework-constrained CREST response projector improves galaxy-held-out SPARC prediction over RAR, radius/ g_{bar} nuisance controls, and no-retuning NFW/Burkert halo-prior predictors. The improvement shows matched null-collapse for all galaxies and for the $Q = 1 + 2$ validation class. Flexible per-galaxy halo fits remain stronger raw curve-fitting ceilings. The coherence-weighted upgrade interprets irreversible loading through $\Theta_{\chi} = \Phi_{\text{irr}}/(\chi\eta^2 I + \epsilon)$, preventing the load sector from being used as generic scatter.

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